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Real-World Effectiveness of Omalizumab Treatment in Adult Asthma Patients

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Purpose: Omalizumab improves clinical outcomes for patients with severe asthma (SA), but its long-term effectiveness and potential biomarkers for predicting patient response require further investigation. This study aimed to evaluate the real-world effectiveness of omalizumab in treating SA and to identify potential biomarkers for predicting a favorable treatment response.

Materials and Methods: Clinical outcomes were compared between asthma patients receiving omalizumab (omalizumab group) and those on inhaled corticosteroid with long-acting beta-agonist (ICS-LABA) alone (ICS-LABA group). Propensity score matching and Cox proportional hazards model were used to calculate hazard ratios (HRs). Study outcomes included severe asthma exacerbation (SAE), incompletely controlled asthma, intravenous (IV) corticosteroid use, and asthma-related hospitalization. Incompletely controlled asthma was defined by blood eosinophil counts ≥150 cells/μL, fractional exhaled nitric oxide (FeNO) ≥25 ppb, forced expiratory volume in one second (FEV1%) <80%, or SAE occurrence.

Results: The omalizumab group had significantly lower risks of SAE (HR 0.17, p=0.03), incompletely controlled asthma (HR 0.56, p=0.04), IV corticosteroid treatment (HR 0.38, p=0.02), and asthma-related hospitalization (HR 0.27, p=0.05). Blood eosinophil count stayed lower in the omalizumab group. FEV1% was higher with the omalizumab group, while blood neutrophil count, FeNO, and serum total IgE showed no differences. Furthermore, subgroup analysis showed patients with treatment-favorable response (>50% reduction in systemic corticosteroid dose) exhibited decreased blood neutrophil counts but increased FEV1% and serum total IgE levels compared with the treatment-unfavorable group.

Conclusion: Omalizumab treatment effectively reduces SAE and improves lung function and asthma control. Blood neutrophil counts and serum total IgE may be potential biomarkers for predicting favorable responses to omalizumab treatment.

Key Words: Asthma, IgE, omalizumab, neutrophils

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INTRODUCTION

Asthma is a chronic respiratory disease characterized by significant heterogeneity due to genetic, environmental, and phenotypical variations, leading to airway inflammation. As inhaled corticosteroids (ICSs) have become the main treatment for patients with asthma, the age-standardized prevalence and mortality rates have decreased from 1990 to 2019. Despite these advancements, asthma continues to affect 262.41 million people worldwide, underscoring the persistent severity of its global burden.[1] Furthermore, recent findings suggest that between 3.5% and 5.4% of asthma patients have severe asthma (SA), necessitating the use of high-dose ICSs

combined with long-acting beta-agonists (ICS-LABAs) for effective symptom control.[2] However, some patients receiving high-dose ICS-LABA therapy may still experience uncontrolled symptoms. For these individuals, the addition of biological treatments, such as monoclonal antibodies targeting specific inflammatory pathways, is considered an adjunctive therapy.

Omalizumab, an anti-IgE monoclonal antibody, represents a critical advancement in biological treatments for chronic urticaria, food allergy, and asthma. Its use over several years has notably improved outcomes for patients with SA, a finding substantiated by numerous controlled studies.[3] Moreover, studies focusing on real-world applications of omalizumab in asthmatic patients have yielded consistent results, emphasizing its practical relevance in clinical settings.[4-6] The efficacy of omalizumab in enhancing asthma management, particularly in reducing asthma exacerbations (AEs) and improving control, has been well documented through controlled trials and recent real-world studies.[4,6] In particular, recent real-world evidence (RWE) has indicated that omalizumab treatment in patients with asthma can reduce oral corticosteroid (OCS) use and improve pulmonary function.[7-9] However, despite these advancements, reliable biomarkers for predicting the RWE of omalizumab treatment in asthmatic patients are lacking.

This study was based on the need for additional research and evaluated the RWE of omalizumab treatment (especially in AE and control status) based on 1-year observations in a real-world setting. Furthermore, it aimed to identify potential biomarkers that could predict favorable or unfavorable responses to omalizumab treatment in patients with asthma.

MATERIALS AND METHODS

This retrospective analysis utilized deidentified patient data from Ajou University Medical Center (AUMC), one of the largest tertiary hospitals in South Korea. This study employed a database in which the electronic health records of more than 3 million patients were transformed into the Observational Medical Outcome Partnership-Common Data Model (OMOP-CDM) version 5.3.4, an international standard for healthcare data. Due to the inability to identify individual patients, the requirement for informed consent was waived. The data used for this research were collected on September 19, 2023. The authors did not have access to any information that could iden-

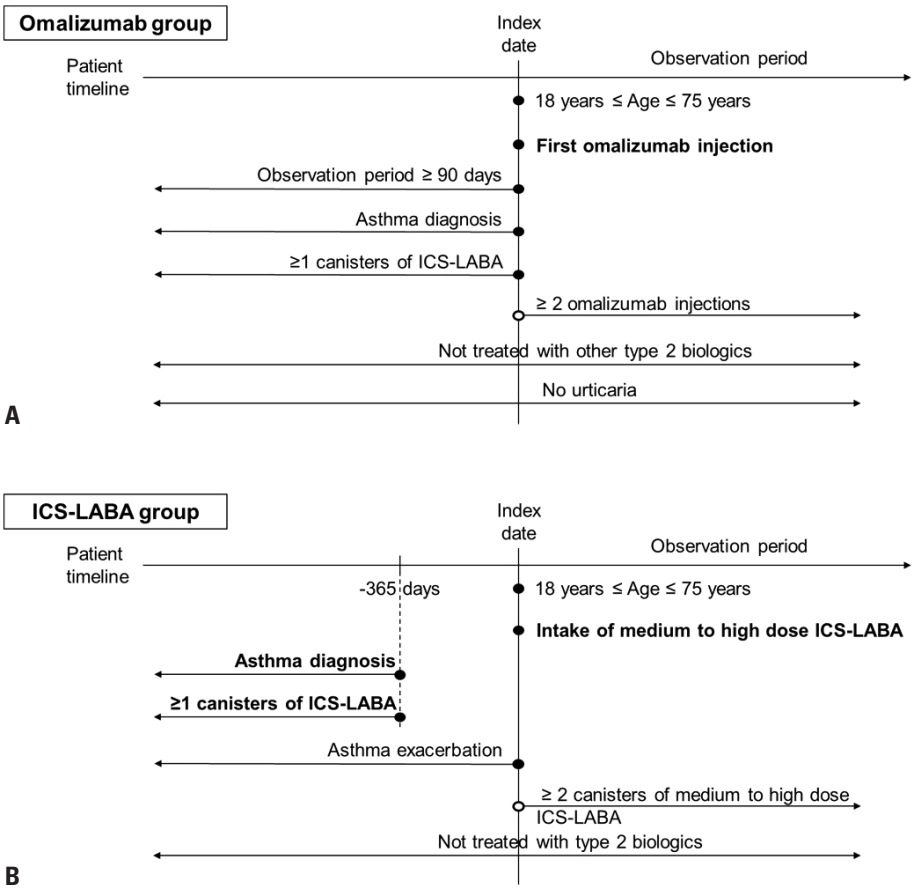

Fig. 1. Definitions of the omalizumab and ICS-LABA groups. (A) Adults with severe asthma from January 1994 to February 2023 who had consistently taken omalizumab. (B) Adult patients with severe asthma from January 1994 to February 2023 who had asthma for at least 1 year and had taken an ICS-LABA without type 2 biologics. ICS-LABA, inhaled corticosteroid with long-acting beta-agonist.

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tify individual participants during or after the data collection process. This study was approved by the Institutional Review Board of AUMC (IRB No: AJOUIRB-DB-2023-462) and adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines, which aim to enhance the clarity and transparency of epidemiological studies. The study was conducted in accordance with the tenets of the Declaration of Helsinki.

Study population and design

The cohort definitions for the omalizumab and ICS-LABA groups are depicted in Fig. 1. Both cohorts consisted of adult asthma patients aged 18 years to 75 years who visited our institution between January 1994 and February 2023. The omalizumab group comprised patients who received three or more injections of omalizumab and had a prior asthma diagnosis before the index date, which was the first day of omalizumab treatment. Patients lacking a minimum observation period of 90 days before this index date were excluded. Patients who had been treated with other type 2 biologics or diagnosed with urticaria were excluded from the omalizumab group. Similarly, the ICS-LABA group comprised patients who had been prescribed medium- to high-dose ICS-LABAs at least three times after the index date. These individuals were identified as SA patients who had been treated for more than a year and who were characterized by a history of AE. Patients treated with type 2 biologics were excluded from the ICS-LABA group.

Severe asthma exacerbation (SAE) was investigated as a primary outcome, and secondary outcomes included incompletely controlled asthma, intravenous (IV) corticosteroid treatment, and asthma-related hospitalization. Furthermore, the key laboratory values from the index date to the end of follow-up were compared between the two groups. Finally, various sensitivity and subgroup analyses were conducted.

Definitions and statistical analysis

SAE was defined as symptom exacerbation that required the use of systemic corticosteroids (≥15 mg/day of prednisolone or its equivalent for ≥3 consecutive days or IV/intramuscular corticosteroids equivalent to ≥60 mg/day of prednisolone) and resulted in hospitalization or emergency department visits. Additionally, we broadened the definition of AE to include unscheduled clinic visits for asthma symptoms without systemic steroid use. Furthermore, incompletely-controlled asthma was defined by the following criteria: blood eosinophil counts of 150 cells/μL or more, fractional exhaled nitric oxide (FeNO) levels of 25 ppb or more, forced expiratory volume in one second (FEV1) below 80%, or the occurrence of an AE. AE was defined as an event requiring SAE-level steroid treatment or asthma-related hospitalization. Completely controlled asthma refers to a condition where none of the criteria for incompletely controlled asthma are met.

In this study, categorical variables are represented as percentages, whereas continuous variables are shown as the mean± standard deviation (SD). To mitigate bias from confounding factors, propensity score matching (PSM) was employed to compare the two groups. A logistic regression model was used to calculate propensity scores (PS). We then applied a greedy (nearest neighbor) matching technique, adhering to a caliper of 0.2 SDs on the logit scale, as proposed by Austin.[10] The absolute standardized mean difference (aSMD) was utilized to assess the balance of covariate distribution between the two groups. Age in 5-year bands, sex, and certain chronic and allergic conditions observed during the 365 days of the lookback windows were matched during the PSM process. Analysis was conducted in an as-treated setting, where the follow-up for patients in each group was discontinued upon cessation of medication. Follow-up was discontinued in the omalizumab group after 90 days without treatment and in the ICS-LABA group after 30 days without treatment. Incidence rates (IRs) per 1000 person-years (PYs) were calculated. Hazard ratios (HRs) with 95% confidence intervals (CIs) were determined using the Cox proportional hazards model, ensuring that the proportionality assumptions were checked through Schoenfeld residuals. In the Cox model, treatment was the sole covariate. Kaplan–Meier curves and log-rank tests were utilized to calculate the cumulative incidence by group and to assess differences in outcomes between groups. Additionally, a mixed-effects regression analysis was conducted to evaluate the significance of the difference in the change in laboratory values over the treatment period between the omalizumab and ICS-LABA groups. The mixed-effects regression model was adjusted for time, treatment group, age, sex, and interaction between time and group, with the patient's research ID included as a random effect to accommodate repeated measurements.

The sensitivity analysis was performed utilizing another PS adjustment method (PS stratification with five strata) and varying the number of matches (1:1, 1:2, or 1:4). Additionally, within the omalizumab group, a subgroup analysis was conducted to compare the key laboratory values between patients with different corticosteroid response patterns. The omalizumab group was stratified into two subgroups: the treatment-favorable group, defined as patients showing more than 50% reductions in cumulative corticosteroid doses in the year after the index date compared with baseline levels; and the treatment-unfavorable group, comprising those who did not meet this criterion. This subgroup analysis aimed to explore potential biomarkers associated with treatment responses among the omalizumab group.

A p value<0.05 was considered to indicate statistical significance. The analyses were conducted with R software (version 4.2.2), along with the open-source Observational Health Data Sciences and Informatics (OHDSI) CohortMethod R package (version 4.2.3). The Cyclops R package (version 3.1.2) facilitated large-scale analysis.[11,12] Additionally, mixed-effects regression models were created using the lme4 package (version

1.1.31). To enhance the reproducibility and transparency of our analysis, the source code has been made available on GitHub at https://github.com/ABMI/omalizumab_asthma (accessed Mar 01, 2024). This repository contains the code for the mixed-effects regression model and an R package designed for computing various statistics and fitting the Cox proportional hazards model.

RESULTS

Study population

The baseline characteristics and aSMDs between the groups before and after PSM are shown in Table 1. The omalizumab and ICS-LABA groups comprised 121 and 641 patients, respectively, before PSM (Supplementary Fig. 1, only online). Prior to PSM, the mean ages were 46.9±14.4 years and 50.6±16.3 years in the omalizumab and ICS-LABA groups, respectively (aSMD=0.23). The proportion of female was greater in the omalizumab group (65.7%) than in the ICS-LABA group (55.5%; aSMD=0.21). There was a difference in the prevalence of hypertensive disorders, diabetes mellitus, hyperlipidemia, pneumonia, rheumatoid arthritis, and allergic rhinitis between the two groups (all aSMDs >0.1).

Through PSM, 107 patients were equally distributed into the omalizumab and ICS-LABA groups. After matching, the mean ages were 47.0±14.8 years and 44.4±15.8 years in the omalizumab and ICS-LABA groups, respectively, with patients appropriately matched into 5-year interval age groups. Only one segment, the 20- to 24-year-old group, was not perfectly matched (aSMD=0.23) due to dropout during the matching process. Diabetes mellitus, which had the largest difference after match-

Table 1. Comparisons of Baseline Characteristics and Comorbidity Profiles between the Omalizumab and ICS-LABA Groups Before/After PSM

Characteristics	Before PSM			After PSM		
	Omalizumab (n=121)	ICS-LABA (n=641)	aSMD	Omalizumab (n=107)	ICS-LABA (n=107)	aSMD
Age group						
15–19 years	3.7	6.1	0.11	3.7	4.7	0.05
20–24 years	3.0	5.7	0.13	3.7	9.3	0.23
25–29 years	6.7	5.5	0.05	7.5	9.3	0.07
30–34 years	9.0	6.5	0.09	9.3	6.5	0.10
35–39 years	6.7	8.4	0.06	6.5	7.5	0.04
40–44 years	15.7	9.2	0.20	13.1	14.0	0.03
45–49 years	12.7	8.8	0.13	12.1	10.3	0.06
50–54 years	9.0	10.2	0.04	7.5	7.5	0.00
55–59 years	12.7	10.8	0.06	13.1	10.3	0.09
60–64 years	8.2	9.7	0.05	10.3	7.5	0.10
65–69 years	6.0	9.8	0.14	6.5	7.5	0.04
70–74 years	6.0	7.7	0.07	5.6	5.6	0.00
75–79 years	0.7	1.7	0.09	0.0	0.0	0.00
Female, %	65.7	55.5	0.21	60.7	58.9	0.04
Medical history (general)						
Hypertensive disorder	7.5	12.4	0.17	7.5	5.6	0.08
Diabetes mellitus	3.0	6.1	0.15	3.7	1.9	0.11
Hyperlipidemia	6.7	2.4	0.20	6.5	4.7	0.08
Pneumonia	3.0	7.9	0.22	3.7	4.7	0.05
Rheumatoid arthritis	0.7	2.0	0.11	0.9	1.9	0.08
Bronchitis	5.2	3.6	0.08	3.7	3.7	0.00
Medical history (allergic disease)						
Allergic rhinitis	50.0	36.3	0.28	51.4	54.2	0.06
Allergic conjunctivitis	6.7	7.0	0.01	8.4	10.3	0.05
Concomitant drugs						
Medium to high dose of ICS-LABA	89.6	97.9	0.53	92.5	96.3	0.16
IV corticosteroid	67.2	74.9	0.18	72.0	79.4	0.17
Oral corticosteroid	91.8	89.0	0.09	93.4	87.9	0.19
IV or oral corticosteroid	93.3	94.5	0.05	95.3	93.5	0.08

ICS-LABA, inhaled corticosteroid with long-acting beta agonist; PSM, propensity score matching; aSMD, absolute standardized mean difference; IV, intravenous. Data are presented as %.

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ing, had an aSMD of 0.11, and all other comorbidities were appropriately matched (all aSMDs <0.1).

Clinical outcomes

In the omalizumab group, the follow-up period had a median duration of 125 days, totaling 60.9 PYs. In the ICS-LABA group, the follow-up period had a median of 28 days, accounting for 10.3 PYs. The risk of SAEs was significantly lower in the omalizumab group (IR, 147.92/1000 PYs) than in the ICS-LABA group [IR, 1163.22/1000 PYs; HR 0.17 (95% CI 0.03–0.61)] (Table 2). Furthermore, the risk of incompletely controlled asthma was significantly lower in the omalizumab group (IR, 2621.41/1000 PYs) than in the ICS-LABA group [IR, 6010.62/1000 PYs; HR 0.56 (95% CI 0.32–0.95)]. The proportion of patients who showed incompletely controlled asthma control was greater in the ICS-LABA group (71%) than in the omalizumab group (60%) (Fig. 2). Additionally, IV corticosteroid treatment and asthma-related hospitalization outcomes were also significantly lower in the omalizumab group than in the ICS-LABA group (HR=0.38, p=0.02 and HR=0.27, p=0.05, respectively). The log-rank test revealed significant improvements in the risk of SAEs, AEs, and IV corticosteroid treatment (p=0.021, p=0.013, and p<0.01, respectively) (Fig. 2) and a marginal difference in asthma-related hospitalization (p=0.062).

Table 2. Differences in the Incidence and HR of Clinical Outcomes between the Omalizumab Group and the ICS-LABA Group

Outcomes	Omalizumab (n=107)		ICS-LABA (n=107)		HR (95% CI)	*p* value
	n (%)	IR/1000 PYs	n (%)	IR/1000 PYs		
Severe asthma exacerbation	9 (8.4)	147.92	12 (11.2)	1163.22	0.17 (0.03–0.61)	0.03*
Incompletely controlled asthma	69 (64.5)	2621.41	43 (40.2)	6010.62	0.56 (0.32–0.95)	0.04*
IV corticosteroid treatment	22 (20.6)	394.30	22 (20.6)	2635.45	0.38 (0.16–0.83)	0.02*
Asthma-related hospitalization	15 (14)	273.31	11 (10.3)	1082.37	0.27 (0.06–0.87)	0.05*

HR, hazard ratio; ICS-LABA, inhaled corticosteroid with long-acting beta-agonist; IR, incidence rate; PY, person-year; CI, confidence interval; IV, intravenous.
*Statistically significant.

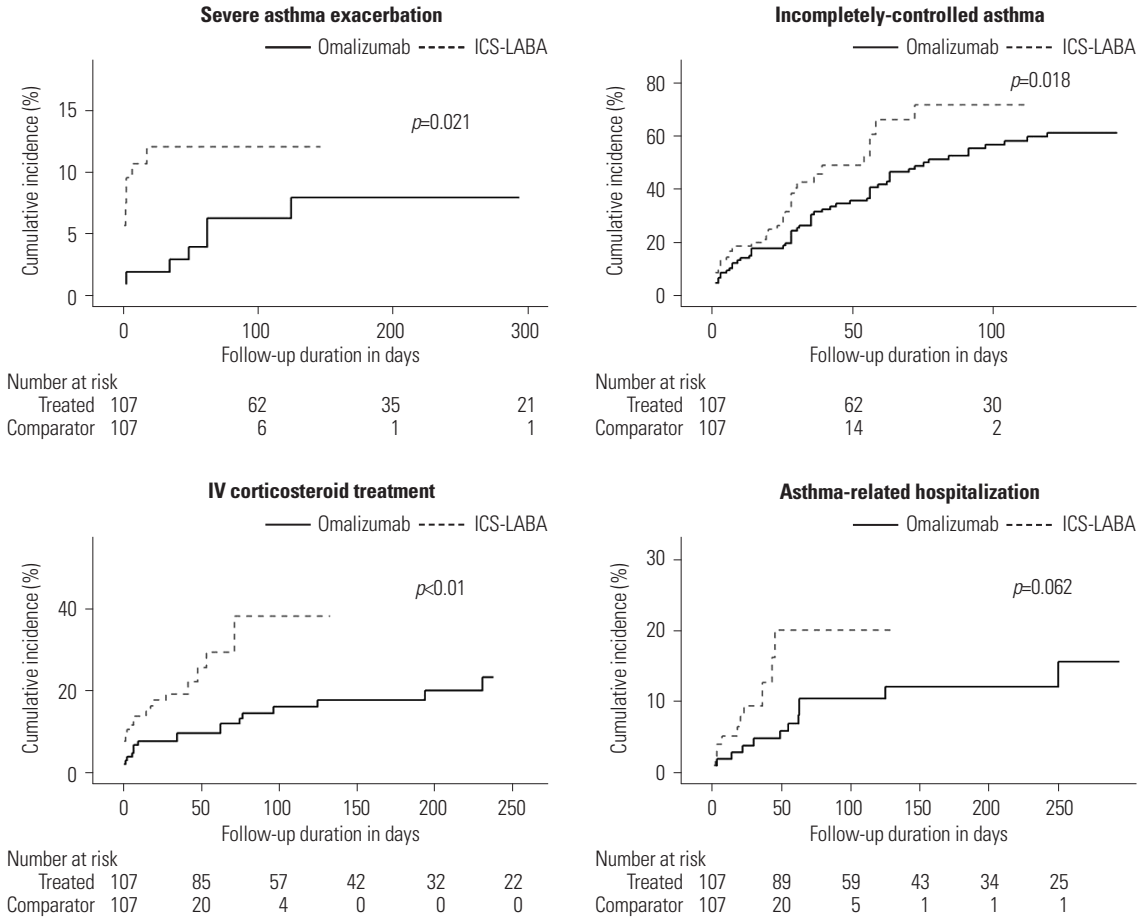

Fig. 2. Comparisons of the cumulative incidence of severe asthma exacerbations and other outcomes between the omalizumab and ICS-LABA groups. *p* values were derived from the log-rank test. ICS-LABA, inhaled corticosteroid with long-acting beta-agonist; IV, intravenous.

Using mixed-effects regression analysis, we assessed the differences in the rate of change in laboratory values between the omalizumab group and the ICS-LABA group. The results revealed significantly higher blood eosinophil and erythrocyte sedimentation rates (ESRs) in the ICS-LABA group than in the omalizumab group (all $p<0.01$) (Supplementary Table 1, only online and Fig. 3). Conversely, the FEV1% was significantly higher ($p=0.02$) in the omalizumab group than in the ICS-LABA group. However, no significant differences were noted in blood neutrophil counts, serum total IgE levels, or FeNO between the groups ($p=0.96$, 0.84, and 0.56, respectively). The results of the group effects at various time points for each laboratory value are shown in Supplementary Table 2 (only online).

Sensitivity analysis

According to the various analyses illustrated in Fig. 4, the risk of SAEs was consistently lower in the omalizumab group than in the ICS-LABA group across all but one analysis (stratification, $p=0.07$). For outcomes related to IV corticosteroid treatment, the omalizumab group had a significantly lower risk in all analyses (all $p<0.05$). There were significant reductions in the risk of incompletely controlled asthma in one setting (1:1 PSM, $p=0.04$) and asthma-related hospitalization in two settings (1:1 PSM and 1:2 PSM; all $p=0.05$).

Subgroup analysis

Comparative analysis of laboratory values between the treatment-favorable and treatment-unfavorable groups revealed that blood neutrophil and leukocyte counts decreased more markedly in the treatment-favorable group than in the unfavorable group ($p=0.02$ and $p=0.03$, respectively) (Supplementary Table 3, only online and Fig. 5). Conversely, FEV1% and serum total IgE levels showed a significant increase in the treatment-favorable group ($p=0.05$ and $p<0.01$, respectively) (Supplementary Table 3, only online), while blood eosinophil

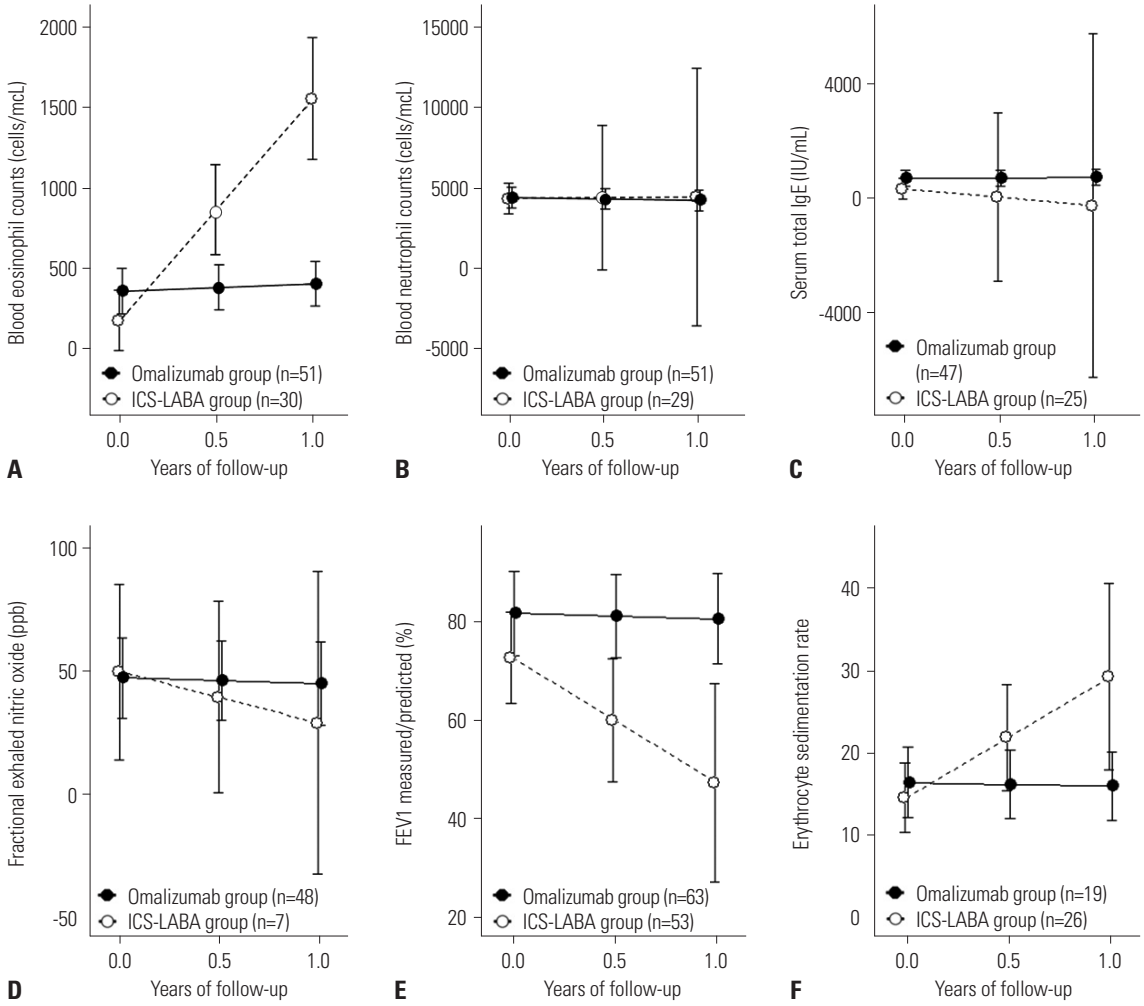

Fig. 3. Longitudinal changes in laboratory values during the follow-up period according to the linear mixed-effects model in the omalizumab (solid line with a closed circle) and ICS-LABA (dashed line with an open circle) groups. (A) Blood eosinophil counts (cells/μL). (B) Blood neutrophil counts (cells/μL). (C) Serum total IgE (IU/mL). (D) Fractional exhaled nitric oxide (ppb). (E) FEV1 measured/predicted (%). (F) Erythrocyte sedimentation rate. Error bars indicate a 95% confidence interval. ICS-LABA, inhaled corticosteroid with long-acting beta-agonist; FEV1, forced expiratory volume in one second.

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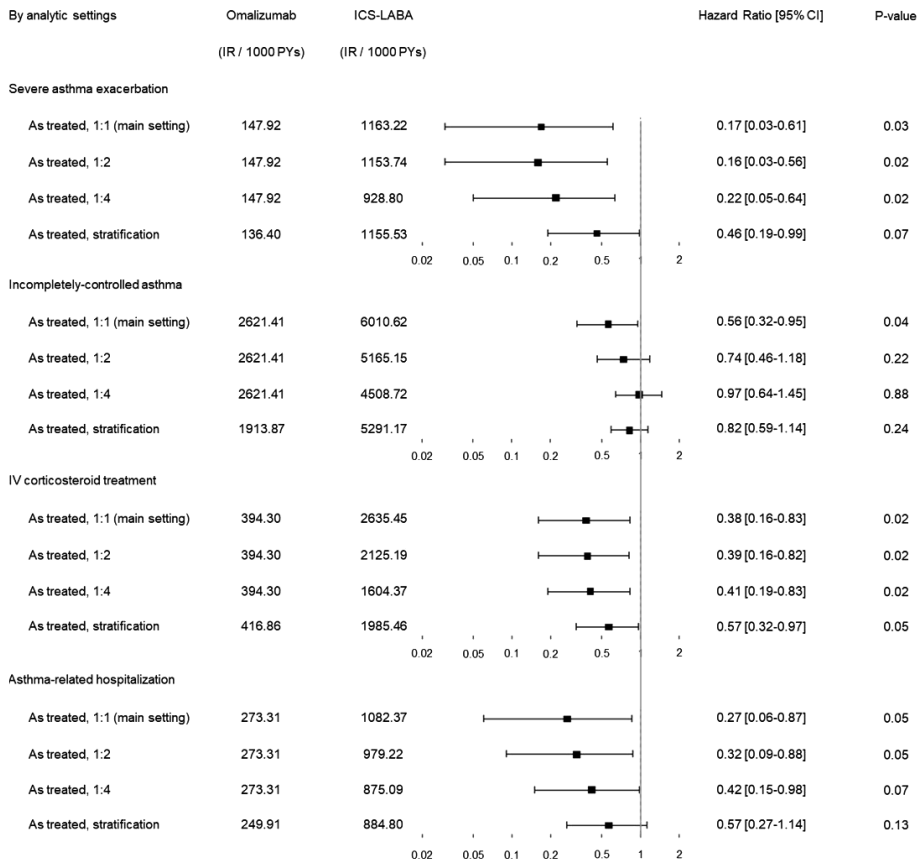

Fig. 4. Forest plot of the sensitivity analyses for severe asthma exacerbations and other outcomes based on various analytic settings. ICS-LABA, inhaled corticosteroid with long-acting beta-agonist; IR, incidence rate; PY, person-year; IV, intravenous; CI, confidence interval.

counts and FeNO levels did not exhibit significant differences between the groups (p=0.60 and p=0.23, respectively). Finally, although there were no significant differences in blood neutrophil or leukocyte counts between the groups at the index date, these parameters were significantly lower in the treatment-favorable group than in the treatment-unfavorable group during follow-ups of 1 year or longer (Supplementary Table 4, only online).

DISCUSSION

Previous studies have shown that omalizumab is beneficial for improving asthma-related outcomes.[7,8,13] Controlled trials have consistently demonstrated the efficacy of omalizumab in enhancing asthma control, reducing AEs, and improving patient-reported outcomes.[3,14] Real-world studies further corroborated the efficacy of omalizumab in improving asthma control and reducing AEs in diverse patient populations but were limited by shorter observation periods (1–2 years).[5,6] Another systemic review showed that long-term treatment with omalizumab (for up to 4 years) significantly improved patient outcomes.[9] The present study confirmed that additional treatment with omalizumab effectively reduced the frequency of AEs in patients with uncontrolled asthma and achieved better control. In addition, this treatment effectively reduced inflammatory markers (blood eosinophils and leukocytes) compared to treatment with ICS-LABAs. Moreover, half of the patients in the omalizumab group were identified as favorable responders, showing a marked reduction in the cumulative dose of corticosteroids per year, which was comparable to the findings of a previous study that showed a 75% reduction in AEs and a 30% reduction in OCS use after 2 years of omalizumab treatment.[15] Furthermore, blood neutrophil counts decreased in the treatment-favorable group compared to the unfavorable group. This study underscores the long-term advantages of sustained omalizumab treatment, such as reduced AEs, hospitalizations, and OCS use; improved lung function; and maintenance of clinical remission status. The results of a systematic review and randomized controlled trials further support these findings.[15,16] Although a prior study in the United States indicated that omalizumab may be cost-effective, further assessments integrating real-world data on factors such as discontinuation and asthma-related outcomes are needed to understand the value of omalizumab.[17] Taken together, the findings of the present study demonstrated that long-term treatment with omalizumab is effective for asthma control (in terms of fewer AEs/asthma-related hospitalizations), improves the FEV1%, and can

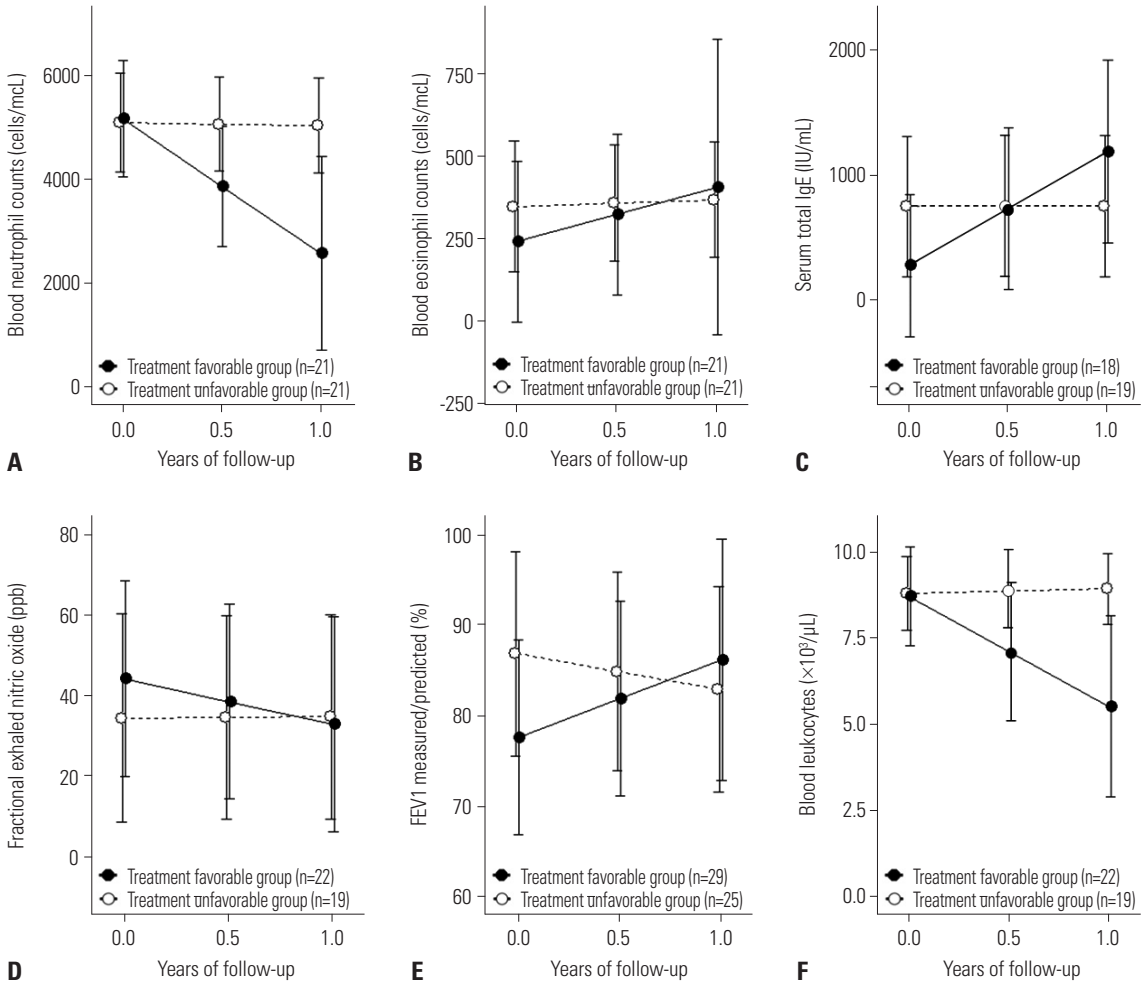

Fig. 5. Longitudinal changes in laboratory values during the follow-up period according to the linear mixed-effects model in the treatment-favorable (solid line with a closed circle) and treatment-unfavorable (dashed line with an open circle) subgroups within the omalizumab group. (A) Blood neutrophil counts (cells/μL). (B) Blood eosinophil counts (cells/μL). (C) Serum total IgE (IU/mL). (D) Fractional exhaled nitric oxide (ppb). (E) FEV1 measured/predicted (%). (F) Blood leukocytes (×10³/μL). Error bars indicate the 95% confidence interval. ICS-LABA, inhaled corticosteroid with long-acting beta-agonist; FEV1, forced expiratory volume in one second.

suppress blood eosinophils more effectively than the ICS-LABA group in patients with asthma in real-world practice.

In recent years, clinical remission has emerged as a potential target in the chronic management of adult asthma. Clinical remission encompasses a well-controlled asthma status characterized by the control of symptoms, exacerbations, and need for OCSs, representing optimal control of the disease while maintaining anti-asthmatic medications.[18-21] The clinical remission rate in adults with asthma is between 2% and 52%,[18,21] with factors such as blood eosinophil counts, asthma control status, lung function, and prior exacerbation rates associated with achieving clinical remission.[20] The ability of biologics to target specific pathogenic mechanisms of asthma has increased the likelihood of achieving clinical remission. In the present study, a novel outcome, termed incompletely controlled asthma, was identified. The prevalence of incompletely controlled asthma was greater in the ICS-LABA group than in the omalizumab group; this finding indicates that the prevalence of completely controlled asthma was 40% in the omalizumab group and 29% in the ICS-LABA group, suggesting that omalizumab could help asthma patients maintain clinical remission status. Studies have reported varying remission rates among patients treated with different biologics[19,22,23]: approximately 30% for mepolizumab,[19] 48% for benralizumab,[22] and 47% for dupilumab.[23] One study showed a 22.8% clinical remission rate with omalizumab,[24] which was lower than that in the present study. This discrepancy can be attributed to the fact that the previous study utilized the Asthma Control Questionnaire score to define the level of asthma symptom control. Overall, the present study provides RWE data that long-term treatment with omalizumab could increase the clinical remission rate by up to 40%.

In the present study, changes in blood eosinophil/neutrophil counts between the two treatment groups were compared. Blood eosinophil count remained stable in the omalizumab group, while it increased progressively in the ICS-LABA group; no difference was found in blood neutrophil count. Eosinophilic

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inflammation is associated with poor lung function, frequent AEs, and SA.[25,26] Eosinophil counts are considered a prognostic factor and have been monitored as a biomarker for assessing treatment response in asthmatic patients using biologics.[27] Other biologics that inhibit interleukin-5 and decrease eosinophilic inflammation have been well reported. Omalizumab does not directly target eosinophils; however, there has been a lack of thorough evaluation of eosinophilic outcomes. Nevertheless, a recent real-world study reported a 14% decrease in blood eosinophil counts after 1 year of omalizumab treatment in patients with SA.[28] In addition, the ESR (a systemic inflammatory parameter) remained stable compared in the omalizumab group compared to the ICS-LABA group, suggesting that long-term omalizumab treatment improves various asthma-related outcomes and suppresses systemic inflammation, including ESR. Taken together, these findings indicate that omalizumab may indirectly affect eosinophilic inflammation, contributing to its efficacy in achieving better control.

The present study demonstrated that FEV1% remained stable in the omalizumab group while it progressively decreased in the ICS-LABA group. Frequent AEs were closely associated with lung function declines.[29] A lower FEV1% was a risk factor for AEs. These findings suggest that long-term treatment with omalizumab could maintain the FEV1% (over 80%), preventing AE and lung function decline.

In the present study, changes in laboratory parameters were compared between the treatment-favorable and treatment-unfavorable groups among omalizumab-treated subjects. Blood leukocyte and neutrophil counts decreased in the treatment-favorable group compared to those in the unfavorable group. Some studies have demonstrated a close association between neutrophilic inflammation and poor clinical outcomes (frequent AEs and therapy resistance).[30,31] Patients with SA had higher blood neutrophil counts compared to those with nonsevere asthma;[32] increased neutrophil counts were observed in patients with uncontrolled asthma even on anti-asthmatic medications.[33] In addition, the serum total IgE level increased in the treatment-favorable group. This observation may seem contradictory, as the dosage of omalizumab is typically determined based on the patient's serum total IgE levels and body weight, with higher doses being recommended for patients with higher total IgE levels.[34] The omalizumab dosing table aims to achieve an average serum-free IgE level of 25 ng/mL, although no significant difference was noted in serum-free IgE levels between favorable and unfavorable responders.[34] Treatment response has been observed even in patients with very high total IgE levels.[35] Although FcεRIα has been reported to be present in neutrophils, its impact is believed to be minimal.[36] The reduction in neutrophils observed in the present study is likely attributed to two factors. First, a decrease in OCS usage may lead to a decrease in neutrophil counts. Second, viral infections, which are known to be associated with asthma control status in patients with SA,[37] have been reported to decrease following omalizumab administration.[38] This observation suggests that omalizumab may help reduce the frequency of respiratory infections in these patients. Therefore, the reduction in neutrophils is likely due to the effect of reduced OCS use, and decreased virus infection is reported to be related to asthma control status in patients with SA.[37] Regarding IgE levels, some studies have reported a decrease in IgE levels after omalizumab use,[39] while others have suggested that an increase in IgE during treatment may predict a positive therapeutic response.[40] The role of serum total IgE as a biomarker for predicting treatment response to omalizumab requires further investigation. Taken together, these findings indicate that decreased blood neutrophil counts may be a potential biomarker for predicting favorable responses to omalizumab in asthmatic patients. Increased serum total IgE levels during omalizumab treatment (compared to baseline levels) may be a relevant predictor of treatment response.

The present study has some limitations. First, as a retrospective study from a single center, the number of patients included was limited, and the follow-up period was not sufficient. To overcome this limitation, the patient cohort was strictly defined, and a rigorous patient-matching process was performed. Therefore, some observation metrics could not be analyzed over a sufficient time compared to the entire observation period. However, by applying strict criteria to exclude patients with SA who were not compliant with omalizumab or ICS-LABA treatment during the observation period, a more accurate real-world study was possible. Second, the study did not analyze the effects of omalizumab according to dosage or administration intervals, which could impact the observed outcomes. However, regardless of these limitations, a more comprehensive assessment of the effectiveness of omalizumab in a real-world setting was possible, even among patients who did not strictly adhere to the prescribed dosing table. Third, comparative analyses with control status based on symptoms or patient-reported outcomes were not performed. However, by comparing various inflammatory markers and pulmonary function test results collected for over 3 years, it was possible to analyze the long-term outcomes of patients with asthma who were closer to clinical remission.

In conclusion, this real-world study demonstrated the 1-year RWE of omalizumab treatment in reducing AEs, maintaining clinical remission status, and improving lung function in adults with asthma. Identifying potential biomarkers for predicting treatment response, such as blood leukocyte/neutrophil counts and serum total IgE levels, may guide personalized treatment strategies. Future studies should focus on identifying the optimal method for omalizumab treatment, establishing reliable biomarkers for predicting treatment responses, and developing strategies to improve patient outcomes.

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